Dongkwan Kim

Principal Security Researcher · AI Security & Red Teaming · Vulnerability Research

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Driving full-chain AI security, with a mission to build the next generation of Red Teams.

Professional Experience

- **Principal Security Researcher**, Microsoft — Atlanta, GA — Apr 2026 – Present
  - Developed and productized **MDASH**, a multi-model agentic system orchestrating 100+ AI agents for vulnerability discovery, validation, and remediation.
  - Building a next-generation agentic system for binary analysis of Windows kernel drivers.
- **Postdoctoral Fellow**, Georgia Tech — Atlanta, GA — Feb 2025 – Apr 2026
  - DARPA AIxCC Winner: Built LLM-based autonomous fuzzing and exploit agents.
  - Led real-world AI offensive-capability evaluations and continued research on applying AI for security, yielding 5 papers (incl. USENIX Security, WOOT, EMNLP, ASE - Distinguished Paper Award).
- **Senior Engineer**, Samsung Security Center, Samsung SDS — Seoul, South Korea — Aug 2022 – Dec 2024
  - Drove Red Team operations across AI systems, IoT devices, Android apps, and kernel-level mitigations.
    - Secured 1B+ users through static/dynamic security reviews across 30+ consumer/enterprise products.
    - Secured prompt-injection chains against RCE, impersonation, and data leakage.
    - Trained 500+ security engineers and presented at 6 industry/academic venues on emerging AI threats.
    - Briefed executive leadership on key security risks and findings through detailed technical reports.
- **Graduate & Postdoctoral Researcher**, KAIST — Daejeon, South Korea — Mar 2014 – Jul 2022
  - Led 8-member research subgroup and conducted security research on emerging systems:
    - Smartphones & Android: Baseband auth bypass, AOSP privilege escalation, VoLTE exploitation.
    - IoT & Embedded Devices: Variant analysis on 1,100+ firmware, finding 20+ 0-days.
    - Cellular Networks: LTE protocol attacks, policy bypass, free data tunneling.
    - Drones & Wearables: Acoustic/EMI attacks, sensor spoofing, wireless analysis.
  - Delivered 27 publications (9 top-tier venues, 1,500+ citations), 10 patents, and 17 funded projects.
  - Discovered 100+ bugs in commercial systems, including Android and smartphone baseband 0-days.
- **Earlier Security Roles**, Pinion Industries & KAIST CERT — South Korea — 2010–2012; 2013–2014
  - Analyzed automotive CAN messages and in-vehicle components, achieving RCE and wiretapping; led KAIST CERT campus-wide audits, including a critical email attack case leading to arrest.

Education

- **Ph.D., School of Electrical Engineering**, KAIST — Daejeon, South Korea — Mar 2016 – Feb 2022
  - Advisor: Yongdae Kim
- **M.S., School of Electrical Engineering**, KAIST — Daejeon, South Korea — Mar 2014 – Feb 2016
  - Advisor: Yongdae Kim
- **B.S., Computer Science**, KAIST — Daejeon, South Korea — Mar 2010 – Feb 2014

Additional Qualifications

- **CTF & Community:** 5x DEF CON finalist; led 20-person team; organized Samsung CTF; won $115K+; delivered 20+ talks; served as a reviewer of top-tier conferences and journals.